

Lecture 14 - Midterm Exam Review & Final Project Discussion

Midterm Exam: **Monday, October 13**

Section	Date	Time	Location	Modality
EEE 4773	Monday, October 13	8:20pm-10:20pm	Williamson Hall 0100	in-person
EEL 5840 on-campus	Monday, October 13	8:20pm-10:20pm	CSE Computer Sciences/En A101	in-person
EEL 5840 EDGE/Online	Sunday-Monday, October 12-13	anytime	HonorLock	online

Midterm Exam Coverage

- **Coverage:** lectures 1-14 (modules 1-4)
- **Practice exam:** available in the [Assignment-Solutions repo](#)
 - I encourage to solve exercises prior to checking solutions.
 - HW2 solutions will be posted on Saturday 10/11 morning.
- **Allowed Material**
 1. A 1-page letter-sized formula sheet (front and back, handwritten with **pen** or typed). **Formulas only!**
 - **Do not include** pseudo-code, solved exercises, lecture derivations or any written definitions.
 2. Scientific calculator.
- **Total time:** 2 hours

Midterm Exam Coverage

The midterm exam will cover all materials from Lecture 1-14. These include:

1. **Introduction to Machine Learning ([Lectures 1-3](#))**
 - Definition of Machine Learning, Artificial Intelligence and Deep Learning
 - Types of learning in Machine Learning
 - Supervised Learning diagram

- (Linear) Regression

2. Experimental Design and Analysis (Lectures 4-5)

- Feature representation: polynomial basis function, radial basis function, etc.
 - Basis functions in general
- Model selection
- Occam's Razor
- Generalization
- Regularization: ridge and lasso
- Performance Metrics for regression: error metrics, Q-Q plot
- Cross-Validation
- The No Free Lunch Theorem
- Experimental Design
- Hyperparameters tuning
- The Curse of Dimensionality

3. Bayesian Learning (Lectures 5-8)

- Frequentist vs Bayesian statistics
- Bayesian interpretation of Regression Least Squares Objective Function
- Maximum Likelihood Estimation (MLE)
- Maximum A Posteriori (MAP)
- Bayesian Prior Equivalence
- Conjugate Priors, Online update

4. Generative Models (Lectures 9-14)

- Distinction between classification and regression
- Probabilistic Generative Models
- Naive Bayes Classifier
- Mixture Models
 - Gaussian Mixture Models (GMMs)
- The Expectation-Maximization (EM) algorithm
- Hidden latent variables
- K-Means Clustering
- (Fuzzy C-Means and Possibilistic C-Means will **not** be included in the midterm)

How to prepare for exam

This is a suggestion only.

1. Review/read all Notebooks.

2. Create your formula sheet. **Do not include written descriptions, pseudo-code or solved exercises (including derivations).**
 3. Review/redo exercises from HW1 and HW2.
 4. Review/redo exercises from quizzes 1 and 2.
 5. Solve practice exam/s.
 6. Find more practice questions in textbooks.
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Discussion Board Questions

Thank you for posting questions!

Post 1

Topic: Naive Bayes Classifier

Examining the Naive Bayes classifier, the steps involve selecting a distribution for each class, determining the parameters for each model using MAP or MLE, and then calculating the posterior probability of the class. I understand how to perform the MLE, but I'm not sure about the MAP approach. For this one, should I assume a prior probability of the model's probability parameters as well in step one, or how does this work? Since we have a fixed probability for each class, by using MAP, are we having two types of prior, one for the parameters that are uncertain and the other for the labels (classes) that have a fixed value?

In the implementation, I noticed that in the code, we are just using the data likelihood and the prior probability to make predictions. Is this because the evidence is just a constant that can be disregarded? - D. Correa Caselles

Post 2

Will there be programming questions on this exam? - J. Hu

No. The exam will be purely theoretical.

Final Project Topics

Final Project brief description in [Canvas](#).

A poll will open on Thursday, October 9th at 8:00PM EST. One team member must select one project topic. Topics are on a first come, first served basis.
